**CardioClarity Studio**

| | |
|---|---|
| Live demo | cardioclarity-962866578386.us-central1.run.app |
| Code | github.com/ashishmokalkar/Multimodal-AI-Hackathon-Devpost |
| Demo video | youtu.be/oskivSGjGLw |

MULTIMODAL AI HACKATHON 2026 · TRACK A · TEAM: ASHISH MOKALKAR · FAIZAN SHAIKH

# Coronary risk you can see on the heart, with a reason behind every number

**The problem.** Heart disease is India's biggest killer: **2.8 million deaths a year, 28% of all deaths** (GBD India, *Lancet Global Health* 2018), and **52% of them before age 70**, against 23% in the West (Prabhakaran et al., *Circulation* 2016). India has **fewer than five cardiologists per million people** (*JSCAI* 2026), a typical doctor's visit lasts **about two minutes** (Irving et al., *BMJ Open* 2017), and reports are in English, which only **one in ten Indians** speaks (Census 2011). At best, patients leave with one risk number. It answers none of their four questions:

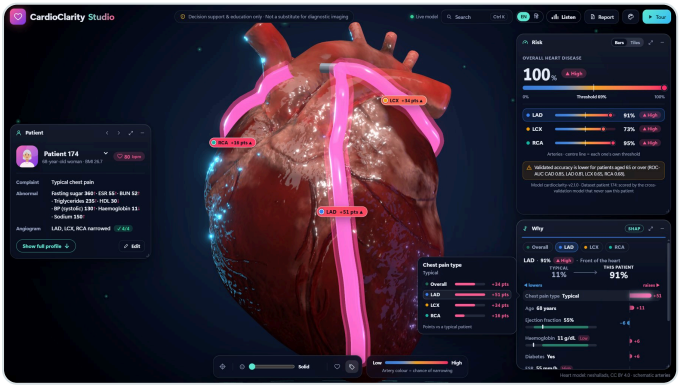
| THE QUESTION | WHAT THEY GET TODAY | WHAT CARDIOCLARITY STUDIO GIVES |
|---|---|---|
| **Where is the problem?** | One number for the heart | The risk of heart disease **and of each main artery**, on a 3D heart |
| **Why me?** | A list of H and L flags | How much **each test result** raised or lowered the risk (SHAP) |
| **What do I do next?** | “Clinical correlation advised” | **Tests to ask about, when to get help**, what-if sliders |
| **In my language?** | English only | A spoken summary in English or **Hindi** (Sarvam AI) |

**Our solution is multimodal at both ends. Seven kinds of clinical data in:** body measures, history, symptoms, examination, ECG, blood tests and echo. **Three ways out:** a 3D picture for the doctor, plain words for the patient, Hindi speech. **Designed next:** vision AI for report photos, voice input, ABHA import with consent (93.95 crore IDs).

## 00 How CardioClarity meets the Track A evaluation criteria

| CRITERION | WHAT WE BUILT | PROOF YOU CAN CHECK |
|---|---|---|
| **Predictive performance** 30% · pages 2–3 | One model for overall heart disease and one for each artery, each picked from more than 10 model types. **Validation:** half the test rounds were used to build it; the other half stayed sealed until the final scores. | **ROC-AUC: CAD 0.939, LAD 0.869, LCX 0.782, RCA 0.770** (1.0 = perfect, 0.5 = guessing). |
| **3D visualization** 25% · page 4 | A realistic 3D heart. **Spatial risk mapping:** each artery changes colour, thickness and glow with its risk. Click, zoom or see through it. | **Smooth on an ordinary laptop: 56–59 fps, no graphics card** (measured). |
| **Clinical interpretability** 20% · page 5 | **SHAP** shows how much each test result raised or lowered the risk. Plus what-if sliders, a plain-language report and a spoken summary in English and **Hindi (Sarvam AI)**. | Press “Explain this prediction”. The points add up exactly to the result. |
| **System integration** 15% · page 6 | The model and the screen share one checked data format. **Real-time updates:** change a value and the heart updates. | **About 1 second** from moving a slider to the heart changing (measured on the live site). |
| **Technical implementation** 10% · page 6 | Runs online on Google Cloud without a graphics card. **Reproducibility:** one command rebuilds the models from the public data. | 70 + 163 automated tests pass; no accessibility errors found. |

**For decision support and education only. Not a substitute for diagnostic imaging.** On every screen and report page.




**Patient 174**, 68. Her angiogram showed all three arteries narrowed, and the app got all four answers right. The Why panel lists what drove her front-artery (LAD) result of 91%. Pointing at “chest pain type” shows how much it raised each artery's risk, on the heart too.

## 01 Predictive performance: the data and how we tested 30%

We used a public dataset of **303 patients** from one heart hospital in Iran (Z-Alizadeh Sani, UCI #411). Every patient had an **angiogram**, the X-ray test that shows narrowed arteries, so the true answer is known. An artery counts as diseased if it is at least 50% narrowed.

| ANSWER | PATIENTS | SHARE | PART OF THE HEART IT FEEDS |
|---|---|---|---|
| **CAD** | 216 | 71% | Any coronary artery disease |
| **LAD** | 177 | 58% | Front of the heart |
| **LCX** | 119 | 39% | Side and back |
| **RCA** | 114 | 38% | Right side and bottom |

Ages 30–86; 127 women. As 71% have CAD, a model that always says “disease” would already be 71% accurate. So we judge models mainly by **ROC-AUC**: how well they separate sick from healthy patients (1.0 is perfect, 0.5 is guessing).

### What goes in (49 inputs)

| | |
|---|---|
| `Body` | Age, sex, weight, height (BMI worked out) |
| `History` | Diabetes, high blood pressure, high cholesterol, smoking, family history, kidney disease, stroke, heart failure, lung and thyroid disease |
| `Examination` | Blood pressure, pulse, swollen legs, weak pulses, lung sounds, heart murmurs |
| `Symptoms` | Chest pain type, breathlessness, activity limits, chest pain on light effort |
| `ECG` | Seven findings on the heart tracing (such as ST and T-wave changes) |
| `Blood tests` | Sugar, kidney tests, cholesterol and fats, inflammation, haemoglobin, salts, blood count |
| `Echo` | Heart ultrasound: pumping strength, wall movement, valve disease |

### Preparing the data (cleaning)

The data had no gaps and no duplicate patients. We:

- **Kept the answers out of the inputs.** The angiogram results (LAD, LCX, RCA) are never used to make a prediction, as the brief requires.
- **Removed a column that was the same for everyone** (chest pain on exertion).
- **Removed columns that repeat others**, such as “obesity”, which is just BMI of 25 or more.
- **Combined related columns:** three chest-pain columns into one, two smoking columns into one.
- **Shuffled patients before every split**, because the file lists most sick patients first.

### Adding useful inputs (feature engineering)

We added a new input only if it made the models better overall, and never clearly worse.

- **Pre-test probability:** a standard doctor's estimate from age, sex and chest-pain type. The single most useful input.
- **Ten published heart-risk scores** (such as Duke and ESC), kept where they helped.
- **A wall-movement flag:** all 28 patients with poor movement in three or more areas of the heart wall had LAD disease.
- **Each model picks its own best inputs** (44 to 66 of them).

### How we tested (validation methodology)

- **Train on some patients, test on others.** We split the patients into 5 groups, trained on 4 and tested on the 5th, and repeated this with 10 different shuffles (repeated cross-validation).
- **A sealed final test.** Half the shuffles were used for every design choice. The other half stayed sealed and were opened once, for the final scores. This stops us from fooling ourselves.
- **No peeking.** Everything learned from data (input selection, scaling, calibration, alert levels) was learned only from the training patients in each round.
- **Real improvements only.** A change counted only if its gain was statistically significant. The ranges shown (95% confidence) come from resampling patients.

### What we tried

More than 10 model types, from simple logistic regression to boosted decision trees (XGBoost, CatBoost); TabPFN, a pre-trained model for tables; AutoGluon's automatic search; models that predict all arteries together; and adding a second hospital's patients (AngioCAD). Each answer kept its own best model.

### What the data showed

- **Chest pain type and age** matter most for every artery.
- **LAD disease shows on the echo** (weak pumping, poor wall movement), as the LAD feeds the front wall.
- **RCA** is linked to inflammation in the blood; **LCX** to age and blood pressure, and is often silent on the ECG.

## 01 Predictive performance: the models and evaluation results 30%

### The four models (model architecture)

| ANSWER | MODEL | INPUTS |
|---|---|---|
| **CAD** | Explainable Boosting Machine (a transparent model) | 45 |
| **LAD** | CatBoost (boosted decision trees) | 53 |
| **LCX** | XGBoost (boosted decision trees) | 66 |
| **RCA** | A blend of 8 simpler models | 44 |

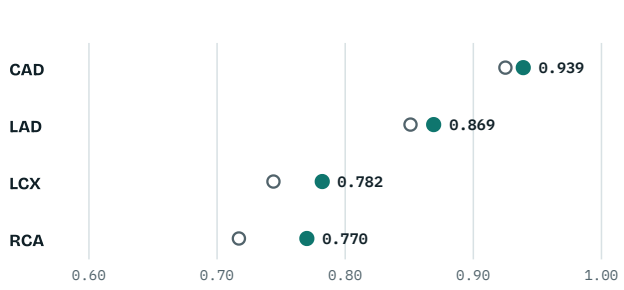
### After the model

- **Calibration:** adjusts the scores so that “70%” really means about 7 in 10 such patients have the disease.
- **Consistency:** an artery's risk is never shown higher than the overall heart-disease risk.
- **Alert levels:** each answer has its own; for LCX and RCA it is set lower, to miss fewer sick patients.
- **Uncertainty:** when the model cannot decide, the app says “unsure” instead of guessing.

### Evaluation results on the sealed test (exactly what the live app does)

| ANSWER | ROC-AUC (95% RANGE) | ACCURACY | BALANCED ACC. | PRECISION | RECALL | F1 | BRIER | ECE |
|---|---|---|---|---|---|---|---|---|
| **CAD** | **0.939** (0.903–0.960) | 0.867 | 0.868 | 0.944 | 0.867 | 0.903 | 0.088 | 0.040 |
| **LAD** | **0.869** (0.822–0.906) | 0.792 | 0.780 | 0.807 | 0.850 | 0.826 | 0.144 | 0.054 |
| **LCX** | **0.782** (0.722–0.820) | 0.673 | 0.698 | 0.561 | 0.812 | 0.660 | 0.191 | 0.052 |
| **RCA** | **0.770** (0.714–0.813) | 0.679 | 0.678 | 0.573 | 0.674 | 0.612 | 0.192 | 0.056 |

**Accuracy:** share of correct answers. **Precision:** of the patients flagged, the share truly sick. **Recall:** of the sick patients, the share caught. **F1:** a balance of the two. **Brier, ECE:** how trustworthy the percentages are (lower is better).




The final models beat the best simple starting model on every answer, most for LCX (+0.038) and RCA (+0.053). Against a basic logistic regression, the gains were statistically significant for LAD, LCX and RCA. CAD was already well predicted by simple models (0.925).

Tried but not kept, because none beat the chosen models: TabPFN, AutoGluon, models that predict all arteries together, and adding the second hospital's patients.

### Can the percentages be trusted? (calibration)

| ANSWER | ERROR (ECE), BEFORE → AFTER | SLOPE (1.0 = IDEAL) |
|---|---|---|
| **CAD** | 0.044 → 0.040 | 1.01 → 0.97 |
| **LAD** | 0.075 → 0.047 | 0.70 → 0.94 |
| **LCX** | 0.061 → 0.053 | 1.20 → 0.92 |
| **RCA** | 0.138 → 0.047 | 0.40 → 0.92 |

After calibration (Platt scaling), every answer is close to ideal.

### Where it works best and worst (ROC-AUC)

| GROUP (PATIENTS) | CAD | LAD | LCX | RCA |
|---|---|---|---|---|
| Men (176) | 0.908 | 0.876 | 0.776 | 0.725 |
| Women (127) | 0.965 | 0.844 | 0.746 | 0.805 |
| Age 50–64 (150) | 0.946 | 0.867 | 0.780 | 0.770 |
| **Age 65+ (95)** | 0.848 | 0.807 | 0.651 | 0.677 |
| Diabetic (90) | 0.975 | 0.739 | 0.727 | 0.694 |

Weakest for patients over 65; the app shows them a warning.

#### No patient is scored by a model that has seen them

A model that saw a patient during training already knows their answer, so it would look perfect (AUC 0.99–1.00). So the live app keeps **five models per answer**, each trained without one fifth of the patients, and always scores a study patient with the model that never saw them. Checked through the live app, it agrees with the angiogram for **87%** of patients on CAD, **81%** on LAD and **71%** on LCX and RCA.

#### Knowing when it is unsure

For CAD the model gives one clear answer for 96% of patients. For LCX and RCA it is unsure for about half, and says so. It is good at telling whether a patient has disease, but only slightly better than chance at telling **which** artery among sick patients (AUC 0.55–0.62). So the heart shows the **chance an artery is involved**, not where a blockage is.

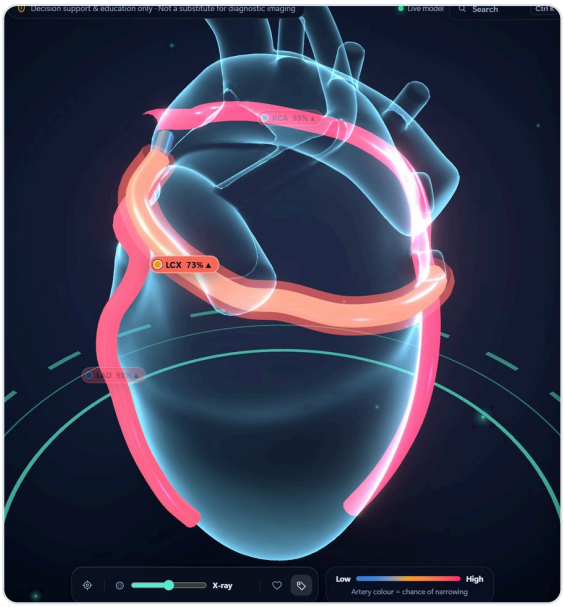
## 02 3D visualization: the heart, risk on each artery, and interaction 25%

### The heart (anatomical representation)

- **A realistic 3D heart model** (“Realistic Human Heart” by neshallads, CC BY 4.0), the same one in the app and in the printed report.
- **The three arteries** (LAD, LCX and RCA) are drawn by us over the heart's surface, each as its own object. They show which area of the heart a result refers to, not the patient's own arteries.

### Risk shown on each artery (spatial risk mapping)

- **Colour:** blue (low) → amber (at that artery's alert level) → red (high). The same colour key is used everywhere.
- **Higher risk also makes an artery thicker and brighter**, with faster blood flow. Arteries above the alert level pulse with the heartbeat, which beats at the patient's own pulse rate.
- **A label on each artery** shows its percentage, ▲ when above the alert level, and “?” when the model is unsure. Point at a test result and the labels show its effect on each artery.
- **Smooth changes:** when a what-if changes the result, the artery slowly changes colour, so the change is easy to see.




**See-through (X-ray) view, LCX selected** (patient 174, 73%). Clicking an artery highlights it, moves the camera to it and opens its explanation.

### Interactive visualization

**Explore**
Rotate and zoom. Slide from a solid heart to see-through to arteries only. Jump to any artery and back.

**Connect**
Click an artery, on the heart or in the Risk panel, to explain it. Point at a test result to see its effect on all three arteries.

**Guide**
A guided tour walks through the result, artery by artery. Ctrl+K searches patients, tests and views.

### From prediction to picture

SERVER
**One result**
The risk for CAD and each artery, the alert levels, and the reasons.

▶

ARTERY MAP
**Match**
Each answer is tied to exactly one artery, by name.

▶

COLOUR KEY
**Translate**
Risk becomes colour, thickness, glow and flow.

▶

BROWSER
**Draw**
The 3D scene is drawn in the browser with Three.js.

▶

REPORT
**Print**
The same scene makes still pictures for the PDF report.

### The colour and the number come from one result

The heart and every panel read the same result, and each artery is matched by name, not by position, so an artery's colour and its number are drawn from the same values. Every result is checked before anything is drawn.

**Accessible:** screen readers get a text summary of the heart, and all motion can be switched off.

### Smooth on an ordinary laptop (measured)

| GRAPHICS | BEATING | ROTATING | FRAME TIME |
|---|---|---|---|
| **Built-in Intel graphics** | **59 fps** | **56 fps** | ≤ 18.5 ms |
| No graphics hardware (CPU only) | 9 fps | 9 fps | ≤ 200 ms |

fps: frames per second (60 is fully smooth). Measured in Microsoft Edge, 1440 × 900 window, over 8 seconds; frame time is for 95% of frames. If frames drop, the app switches to a lighter setting.

## 03 Clinical interpretability: explaining each result with SHAP 20%

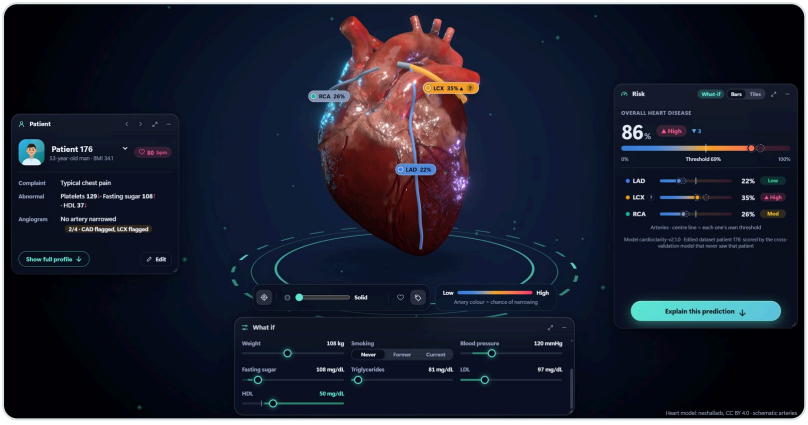
### Why the model said this (feature attribution with SHAP)

For every patient, the app uses **SHAP** (Shapley additive explanations) to show how much each test result pushed the risk up or down, in percentage points, compared with a typical patient. The points add up exactly to the result: if a typical patient's risk is 11% and this patient's is 91%, the points add up to +80. The same numbers appear on the heart and in the Why panel.

We use one SHAP method for all four models, on the final percentage the app shows. We chose SHAP over LIME because its numbers always add up and do not change from one run to the next.

### The breakdown, in the app

- **Why panel:** each test result with its value, normal range, a High/Low flag and its points, biggest first. Click an artery to switch to it.
- **Factor map:** the strongest results across all four answers at once.
- **Measures:** every test against its normal range, filtered to what is abnormal, in mg/dL or mmol/L.
- **Context:** how the patient compares with people of the same age and sex, and what their angiogram found.



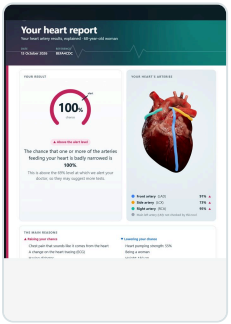

**What-if, patient 176** (53, smoker, clear angiogram). If he stopped smoking and his HDL ("good" cholesterol) rose from 37 to 50, the LCX risk would fall from 51% to 35%, and the overall risk from 89% to 86%. Dotted markers show the original values.

### What-if, in real time

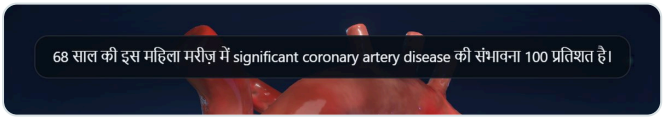
Sliders for things a patient can change (weight, smoking, blood pressure, blood sugar and cholesterol) re-run the model straight away. The real result stays on screen next to the new one. This helps a doctor show a patient why a change matters.

A what-if shows what the model expects, not a promise of what treatment will do.

### Clear for the patient




- **A 3-page report** to print or save as PDF: the risk with the 3D heart and the main reasons; each artery on its own; the patient's numbers against healthy ranges, questions for the doctor, and when to get help.
- **Listen:** a spoken summary with captions: the risk, the arteries, the four main reasons, abnormal readings and the next step.




**Listen in Hindi:** switch EN | हिं next to Listen. The Hindi summaries are recorded with **Sarvam AI** and keep medical words in English, the way Indian doctors speak.

### Simple by design

Medical software often shows everything at once. We show the answer first, and details only when asked.

- **The heart is the screen.** Results sit on the heart, with a few small panels around it, not a wall of charts.
- **Only two panels open at the start** (Patient and Risk). The rest open when needed.
- **The patient summary lists only abnormal results;** the full record is one click away.
- **The start screen** shows four example patients, a search, and the ways to add a new patient.
- **One safety line** on every screen, plain words in the report, light and dark themes, full keyboard use.

### The four example patients against their angiograms

| PATIENT | STORY | CAD | LAD | LCX | RCA | ANGIOGRAM | CORRECT |
|---|---|---|---|---|---|---|---|
| **174** · 68 F | Typical chest pain, diabetes | 100% | 91% | 73% | 95% | LAD, LCX, RCA narrowed | 4/4 |
| **125** · 36 F | Breathless, no chest pain; caught from echo and ECG | 85% | 75% | 14% | 12% | LAD narrowed | 4/4 |
| **151** · 42 M | Typical pain, smoker, family history | 29% | 28% | 29% | 17% | Clear | 4/4 |
| **176** · 53 M | Typical pain in a smoker; the model over-calls | 89% | 28% | 51%? | 29% | Clear | 2/4 |

Each was scored by a model that never saw them; "?" means unsure. For 176, the Why panel shows the high score comes from his chest pain type.

## 04 System integration: how the parts work together 15%

APP
**Send**
The app sends the patient's 49 values to the server, after checking them.

SERVER
**Score**
Each answer is scored by the right model: one that never saw this patient.

SERVER
**Adjust**
Calibrates the percentages, applies the alert levels, marks “unsure”.

SERVER
**Explain**
Works out the SHAP points for all four answers.

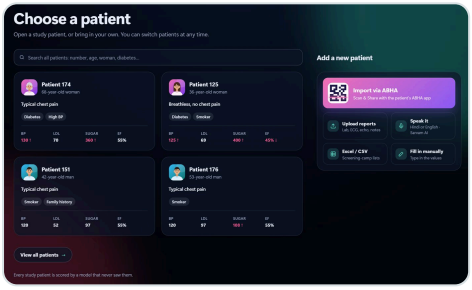
APP
**Show**
The heart and all panels update together from one result.

### Real-time updates: how fast the heart changes (measured)

| FROM MOVING A SLIDER | LIVE SITE | LAPTOP |
|---|---|---|
| Short pause while the slider moves | 300 ms | 300 ms |
| Server: four models plus explanations | 671 ms | 505 ms |
| **Total, typical of 10 tries** | **1.00 s** | **0.84 s** |

### One shared data format

The server (Python) and the app (TypeScript) check every message against the same typed format, so a mismatch shows an error instead of a wrong number. The list of inputs, with units and normal ranges, comes from the server, so the forms and sliders build themselves. Unusual values trigger a warning.




**Start screen**: example patients, search, and five ways to add a patient.

### Getting patient data in, designed for India

Typing 49 values is slow, and most already exist in the patient's records. Every route ends on one review screen, so nothing is scored until a person checks the values.

- `LIVE` **Fill in manually**, with units, normal ranges and checks.
- `DESIGNED` **Import via ABHA:** the patient scans the clinic's QR code with the ABHA app and gives consent. Lab results, diagnoses and vital signs arrive in the standard FHIR format and are matched to our 49 inputs.
- `DESIGNED` **Upload reports:** photos, PDFs or WhatsApp forwards of lab, ECG and echo reports, read by **multimodal (vision) AI** that shows where each value came from.
- `DESIGNED` **Speak it** in Hindi or another Indian language; Sarvam AI turns speech into values.
- `DESIGNED` **Excel / CSV** for health camps: score many patients and rank who to refer first.

**Privacy:** the server keeps no patient data; values are scored and forgotten. ABHA import works only with the patient's consent, in line with India's DPDP Act 2023.

## 05 Technical implementation: how it is built and how to run it 10%

- **Architecture:** a Python server (FastAPI) runs the models; the app is React and Three.js. Both ship in one Docker container on Google Cloud Run, using only CPUs.
- **Easy to extend:** new tests, models or arteries can be added without redesigning the app.
- **Code quality:** 70 Python tests and 163 app tests, including a check that no study patient is scored by a model that saw them.
- **To try it:** open the live demo, pick a patient, press “Explain this prediction”, click an artery, try What-if, then Report or Listen. The Tour button shows everything.

```
# Run the app and server in one container
docker build -t cardioclarity .
docker run -p 8080:8080 cardioclarity

# Or run them separately (Python 3.11, Node 24)
pip install -r requirements.txt
uvicorn api.main:app --port 8000
cd frontend && corepack pnpm install
corepack pnpm dev

# Reproducibility: rebuild the models (CPU)
python scripts/download_data.py
python -m ml.export
```

**Limitations.** The models learned from 303 patients at one hospital in Iran and have not yet been tested on patients from other hospitals. They are weaker for patients over 65 and for the LCX and RCA arteries. Indians often get heart disease younger, so the models must be checked on Indian patients before any clinical use. **Credits:** dataset UCI #411 (Alizadehsani et al., CC BY 4.0); heart model “Realistic Human Heart” by neshallads (CC BY 4.0); open-source EBM, CatBoost, XGBoost, LightGBM, scikit-learn, FastAPI, React and Three.js; Hindi voice by Sarvam AI. Full details: `reports/BEST_MODELS.md`.